Department of Computer Science and Engineering

Academic Year 2022 - 2023 (Odd Semester)

Degree, Semester & Branch: III Semester B.E. ECE ‘A’

Course Code & Title: CS3353 & C Programming & Data Structures

Name of the Faculty member (s): Mrs. K. Jeyakarthika, AP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit V / Quick Sort
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TLO11
- **Activity Chosen:** Flash Cards
- **Justification:**
    - Flash card is a card bearing information which is intended to be used as an aid in memorization.
    - In unit V, the students need to understand various sorting techniques such as insertion sort, quick sort, heap sort and merge sort.
    - The Quick sort concept is also included in data structure laboratory exercise, so the students need to understand the sorting concept and its complexity clearly.
    - This activity helps the students to understand the techniques.
- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
    - Two types of Flash cards were prepared.
    - One card containing question and other card containing answers which is shown in Figure 1.
    - Question cards were distributed to certain group of students and answer cards were distributed to few group of students.
    - The question will be asked by one of the students in the class from the card.
    - Students from the other groups in the class should identify correct answer cards for the respective questions.
    - Students who is having question card and answer card have to explain the concept of quicksort in front of the class which is shown in figure 2.
    - This activity helped the students to revise the different sorting techniques in unit V.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO9	PO10	PSO1
CO5	3	3	2	2	2	2	2

PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO4	PO9	PO10	PSO1
	3	3	3	2	2	2	2
Justification for correlation	Needs engineering and mathematical knowledge for understanding sorting concept	Able to identify relevant sorting, searching problem for the given applications	Able to interpret the complex engineering problems with the help of the sorting algorithms	Research-based knowledge will help to identify the suitable sorting and searching algorithms.	This activity makes student to identify sorting technique individually as well as in team.	Able to explain sorting concepts in detail in front of students.	Able to implement different sorting and searching algorithms in IT applications

- **Images / Screenshot of the practice:**

Flash Cards Questions	Flash Cards Answers
Quick Sort	Divide and Conquer Algorithm
Best Case	O(log n)
Average Case	O(n log n)
Worst Case	$O(n^2)$
Pivot Element	Partition the given list of elements

Figure 1: Flash Cards Questions

Figure 2: Flash Cards Answers

Fig 2: Students involvement in Flash Card Activity

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***
  - All the students were actively participated and enjoyed once they identify answered for the concepts correctly.
  - Students explored the knowledge of the various sorting techniques.
- ***Benefit of the practice:***
  - Students can able to perform different sorting techniques for any applications.
  - They can able to identify the corresponding sorting technique for the applications and implement using C programming.
  - This activity helps the students to attend the sorting technique questions in the examination
- ***Challenges faced in implementation:***
  - Few students found it difficult to find the corresponding flash card which containing answer**.**

References:

- https://www.teachingenglish.org.uk/article/using-flash-cards-young-learners
- https://eltexperiences.com/flashcards-in-the-classroom-ten-lesson-ideas/
- https://www.ritrjpm.ac.in/images/computer-science/FlashCards-Venkat-INP.pdf

Signature of Faculty Member **HOD**